

Ahmed Thabet — Engineering Manager, Platform & Infrastructure

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Engineering Manager with 10+ years building decentralized cloud infrastructure, custom Linux OS, and developer platforms — from bare-metal boot to Kubernetes, Terraform, and SDKs at scale. Led multiple teams shipping Zero-OS and the ThreeFold Grid (1000s of nodes, 60+ countries). Hands-on builder in Go, Python, Rust, TypeScript.

Experience

- *Codescalers / ThreeFold Tech* — **Engineering Manager** (2023 – Present) — Cairo (Remote) Own technical strategy, architecture, roadmap, delivery, hiring, and mentoring for multiple teams building the ThreeFold Grid — a peer-to-peer decentralized cloud. Ship full stack from Zero-OS (custom Linux OS) to SDKs, Terraform provider, and dashboards. Drive cross-team planning, code quality, and on-time delivery across Go, Rust, Python, and TypeScript.
- *Codescalers* — **Software Development Team Lead** (2020 – 2023) Led development of Jumpscale cloud framework, 3Bot virtual system administrator, and Virtual Data Center (VDC) platform — managed Kubernetes, S3, Prometheus/Grafana. Owned architecture, hands-on development, and delivery.
- *Codescalers* — **Software Engineer** (2016 – 2020) Designed and built cloud tooling for VM lifecycle, container orchestration, and infra automation (OpenVCloud / libvirt, AYS orchestration, monitoring & health-checks).
- *Otlob.com* — **Data Analyst** (Freelance) Built audio-fingerprinting pipeline over large-scale video datasets to measure marketing campaign reach.
- *Orascom Telecom* — **Web Developer** (Freelance) Built PHP/CodeIgniter app for server-room environmental monitoring, replacing manual Excel workflows.

Selected Projects — Impact & Ownership

Operating Systems & Core Infra

- *Zero-OS* — Lead development of custom Linux OS powering the entire grid (Go). Bare-metal boot, container/VM isolation, storage & networking.
- *ContentHive* — Content-addressed storage inspired by IPFS (Go).
- *0-Disk* — Network Block Device server with pluggable backends (Tarantool, Lua).
- *haven* (2026) — Privacy-first VCS: private branches, signed access policy, encrypted secrets. Single static Go binary.
- *chvm* (2026) — Docker-like CLI running OCI images as hardware-isolated microVMs on Cloud Hypervisor.

Platforms & Developer Experience

- *KubeCloud* — HA Kubernetes on the grid (Go, Vue.js, PostgreSQL).
- *VDC & Marketplace* — Managed K8s + S3 + monitoring (Python, Stellar, Ansible); one-click app marketplace.
- *Terraform Provider* — IaC for grid provisioning (Go).
- *TFRobot / Gridify / Farmerbot* — Fleet deployment, Heroku-like deploys, and farmer power-management automation (Go).
- *AIBroker* — AI gateway aggregating multiple LLM providers with MCP integration (Rust, Go).
- *aiven / aivenv* (2026) — AI-optimized shell venv compressing CLI output for coding agents (40-90% token savings, Go).
- *pirs* (2026) — Rust port of pi agent harness: OpenAI-compatible loop, coding tools, Rhai extensibility, RPC mode.
- *agent-clip* (2026) — Agent as Pinix Clip — agentic loop with memory, tools, and vision.

SDKs & Frameworks

- *Go SDK* — Primary grid SDK (Go); *TypeScript SDK* — web integration (TS).
- *JS-NG / JS-SDK* — Cloud automation framework & ThreeBot app server (Python, Gevent).
- *EWf* — Embeddable workflow framework for resilient stateful multi-step processes (Go).
- *publishingtools* — Wiki/blog publishing framework (Crystal, Kemal, Svelte).

Open Source Highlights (curated · 797 repos · 282 followers)

- *nimdays* (221 stars) — Nim Days book — practical Nim applications & libraries
- *pygundb* (70 stars) — GUN.js server/client in Python
- *python-sonic-client* (61 stars) — Sonic search backend client (Python)
- *nimassets* (63 stars) — Bundle assets into single Nim binary (→ go-bindata)
- *nim-servy* (31 stars) — Micro web framework for Nim
- *tcprouter* — SNI-based TCP router (Go); basis for *tfgateway* + *coredns-redis* CoreDNS plugin
- *raqueeb* — PID 1 replacement / Foreman alternative
- *rs-retryablehttp* (2026) — Retrying HTTP client for Rust (reqwest, backoff/jitter/Retry-After)
- *ewf* — Workflow framework (Go) · *gofarmer* — Cross-platform farm GUI (Go, Fyne)
- Also: pyparsec, python-pcre, nim-redisclient/parser, vscode-capnp, helloED, PIDA plugins, raml2spore — full list at github.com/xmonader

Skills & Technologies

- **Languages:** Go, Python, Rust, TypeScript, C, Nim, Crystal, Bash — polyglot, picks the right tool
- **Infra & Platform:** Linux (custom OS), Kubernetes / K3s, Helm, Terraform/OpenTofu, Ansible, Docker, Cloud Hypervisor / microVMs
- **Observability & Networking:** Prometheus, Grafana, CoreDNS, Nginx/Traefik, TCP/SNI routing, NBD
- **Data:** PostgreSQL, Redis, MySQL, MongoDB, InfluxDB, Tarantool
- **APIs & Protocols:** REST, GraphQL, gRPC, Cap'n Proto, Stellar
- **Frontend:** Vue, Svelte, React
- **AI / Agents:** LLM gateways, MCP servers, agent harnesses (pi/pirs), token-efficient tooling

Publications & Community

- *Nim Days* — Book on practical Nim (221)
- *Python: Extensive Guide · Introduction to Ruby* — early Arabic community guides
- Translations of PyQt4, PyGTK, Mono IronPython, Cairo tutorials into Arabic

Education

- **BSc Electrical Engineering**, Helwan University, Cairo, Egypt
- English: Business proficiency